

From Scans to Baseline

Task 04-01

How a folder of chest CT scans becomes a labelled dataset and a cross-validated classifier, and what changed in the rebuilt baseline. The review's central point was correct: **AUROC 0.9005 does not by itself establish validity** — the validity lives in how the number was produced. Each of those questions now has an audit in the notebook that prints evidence, and one of them found a real defect.

1 Where Tasks 1–3 got us

Task 1

Literature review

Surveyed 2022–2026 work on LIDC-IDRI nodule classification to establish what accuracy and AUC levels already exist, which architectures dominate, and where the gaps are. This sets the bar the baseline has to clear before any novelty claim is meaningful.

Task 2

Environment

PyTorch, torchvision, MONAI, OpenCV, NumPy/Pandas/Matplotlib, verified locally and on Kaggle with a small CNN trained on MNIST. The point was not MNIST; it was proving the GPU path works end to end before depending on it.

Task 3

Dataset exploration

Read DICOM series, visualised slices, printed geometry: 512×512 slices, pixel spacing 0.664–0.781 mm, 2.5 mm slice thickness. It also surfaced the problem that shaped everything after it.

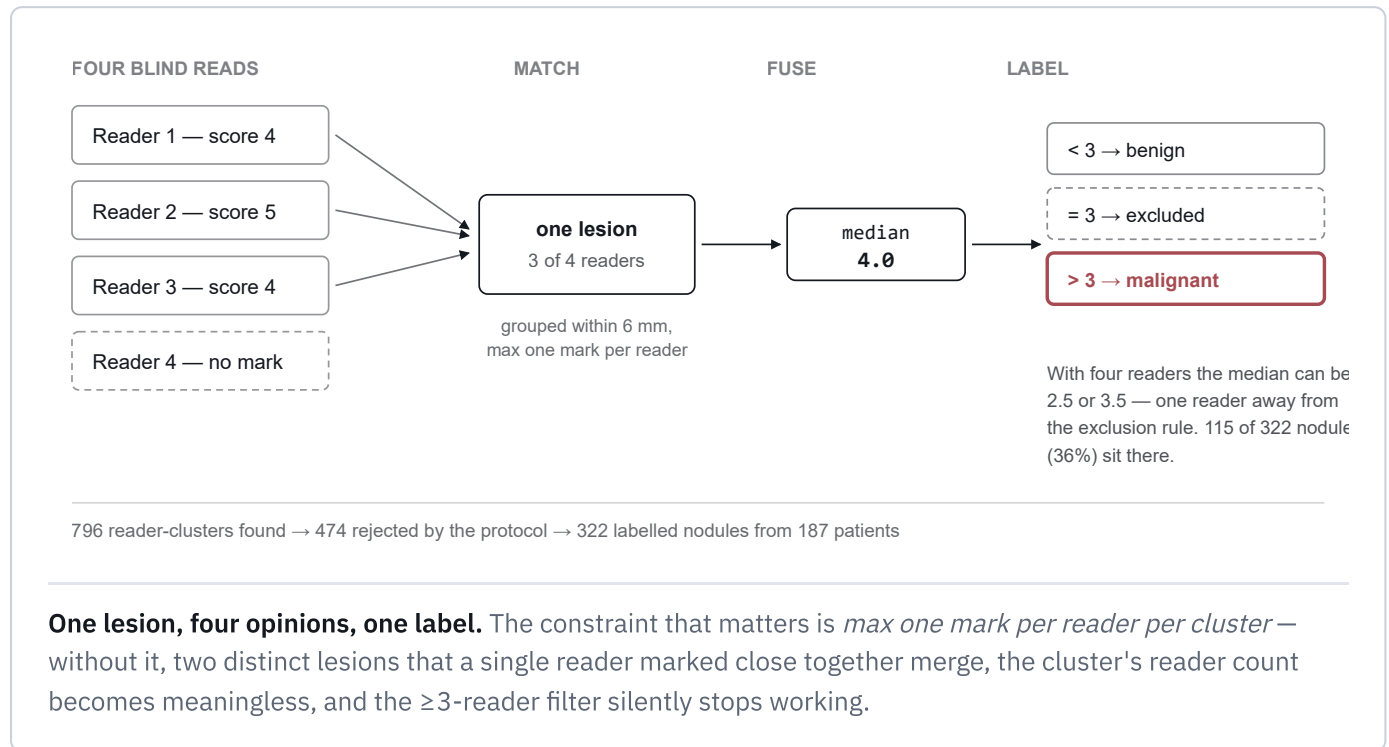
WHAT TASK 3 ACTUALLY FOUND

It printed `no annotations found` for every patient. `tcia_utils` downloads image series only — the LIDC nodule annotations, including the malignancy ratings that *are* the label, ship as a separate XML package.

So there was no supervised task at all until that was fixed, and the largest part of the Task 4 notebook is not the network: it is the pipeline that turns DICOM plus XML into a labelled, cached dataset.

2 How a label is made

LIDC-IDRI has no label column. Up to four radiologists independently outlined every nodule they saw and scored it 1–5 for malignancy. Producing one binary label from four independent opinions takes three decisions, and studies are only comparable when they made them the same way.

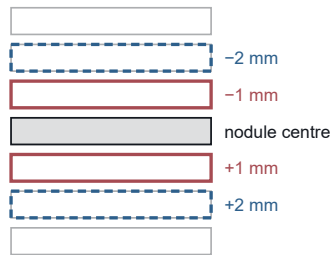


3 The defect the review surfaced

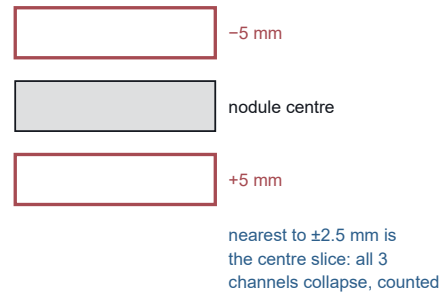
The review asked whether the 3-slice input is centred consistently on the nodule. It was not, and this is the substantive finding of the rebuild.

The *in-plane* crop was already correct: 48 mm of anatomy, converted to pixels using each scan's own `PixelSpacing`. But the two neighbouring slices making up the 2.5D input were taken at ± 1 slice index. LIDC slice thickness ranges roughly 0.6–5 mm, so the through-plane context varied by up to about 8× between patients — and thickness is a property of the acquisition protocol, which varies by institution.

THIN SCAN — 1 mm slices



THICK SCAN — 5 mm slices



OLD: ± 1 slice index

context = 1 mm vs 5 mm

NEW: nearest to ± 2.5 mm

context comparable

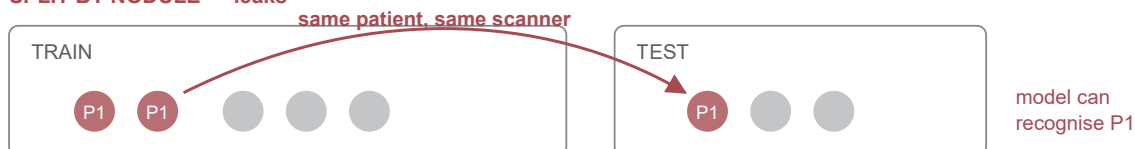
A thick scan simply has no slice at ± 2.5 mm, so the residual cannot be zero — it is measured per nodule and reported, not assumed away.

Same rule, very different anatomy. Under the old rule the through-plane context was a function of the scanner protocol rather than the lesion, in the one dimension nobody was auditing. The new rule cannot make every scan identical — a 5 mm acquisition has no slice at 2.5 mm, so its three channels collapse to one — but that is a property of the data, and the notebook counts how many nodules it affects instead of hiding it. Fixing this changes the pixels, which is why the cache must be rebuilt and the model retrained, and why this is a new notebook rather than an edit.

4 Why the split is at patient level

In this dataset 75 of 187 patients contributed more than one nodule, and one contributed nine. Nodules from one patient share a scanner, a reconstruction kernel and the same anatomy, so splitting on nodules lets a model recognise the patient rather than the pathology.

SPLIT BY NODULE — leaks



GROUP BY PATIENT — clean



What grouping actually prevents. Audit A1 in the notebook does not assert this — it enumerates every fold and every patient and prints the overlap counts: 0 across all five folds and all three split pairs, 0 patients split across test folds, and 322 test predictions covering 322 nodules exactly once.

5 The review's questions, and where each is answered

QUESTION	AUDIT	EVIDENCE PRINTED	VERDICT
Were patient-level folds genuinely separated?	A1	Overlap counts for 5 folds × 3 split pairs; patients split across folds; partition check	PASS
Was preprocessing fitted only on training folds?	A2	Nothing is fitted on data. Sampler weights asserted to reproduce from <code>fold['train']</code> alone	PASS
Did one patient's nodules stay in one fold?	A1.3	75 of 187 patients have >1 nodule; 0 split across folds	PASS
How was the 48 mm patch extracted?	A3	Spacing spread, achieved offsets, patch-overflow count	DOCUMENTED
Is the 3-slice input centred consistently?	A3	Was ± 1 index; now nearest to ± 2.5 mm with residual measured	FIXED
How were malignancy labels constructed?	A4	Build funnel, reader counts, per-rating label mapping	PASS
Did model selection use test/OOF predictions?	A5	Training function takes no test loader (asserted); no hyperparameter search was run	PASS
Do the CIs account for clustering?	A6	Patient-clustered bootstrap, measured against the naive interval	PASS

ON CONFIDENCE INTERVALS

Audit A6 computes both intervals rather than claiming one is better. On this dataset the patient-clustered interval is about **1.2× the width** of the naive nodule-level one — that factor is the error that would have been made by treating 322 nodules as 322 independent observations when they come from only 187 patients.

ON HYPERPARAMETERS

No search was run. Every value was fixed before the first run from ordinary fine-tuning defaults and none was changed in response to a result. That is a stronger guarantee than a tuned model with a carefully separated tuning set, because there is no search over test performance that *could* leak.

6 Reporting honestly

The dataset is **322 nodules from 187 patients**. Every table in the notebook states that n, and nothing is extrapolated to a larger number. At this size the AUROC interval is roughly ± 0.03 , which is wide enough that architecture comparisons may not reach significance — a limitation to state, not to work around.

The false negatives are the number that matters

27 of 181 malignant nodules were missed at threshold 0.5, an FNR of 14.9%. A single figure cannot distinguish "missed the obvious cancers" from "missed the equivocal ones", so the notebook breaks every miss down by the radiologists' own median rating, lesion size and subtlety score. In the previous run that breakdown was reassuring: 15 of the 27 misses were rated 3.5 — the most borderline malignant class — **none at all** were rated 4.5, and the missed lesions were smaller (median 14.2 mm against 23.6 mm) and subtler than the ones caught.

That is a materially different claim from missing obvious cancers, and it is the honest way to describe a 14.9% FNR. Had misses clustered at rating 5, it would have been a safety problem and would need reporting as one.

Two sensitivity analyses

- **Excluding the borderline labels.** 36% of nodules have a median of 2.5 or 3.5, one reader away from exclusion. The models are re-scored without them to show whether the headline rests on cases whose labels are themselves uncertain. This is an evaluation-subset analysis, not a retrain — the two are not conflated.
- **The operating point.** Sensitivity, specificity and false negatives are reported across thresholds, with a clinical threshold chosen on validation rather than left at an

arbitrary 0.5. Which point to deploy at is a clinical decision; the table exists so it can be made from evidence.

Task 04-01 · ResNet-18, patient-level 5-fold CV Notebook: `task04_01_resnet18_cv.ipynb`

n = 322 nodules, 187 patients